Hourly PM2.5 Prediction from Sparse Low-Cost Sensor Records in Uganda: Lessons on Validation, Imputation and Deployment

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Abstract: Low-cost sensor networks have widened PM2.5 monitoring in Ugandan cities, but intermittent device availability leaves short, fragmented records. This paper asks what such records can support. From 14,600 hourly records of 149 AirQo sensors, with readings on only eight days, we built a prediction pipeline, compared twelve models under three validation designs and tested them on a later full month. A first model scored a mean absolute error (MAE) of 3.10 µg/m³ only because it used PM10, a same-hour co-measurement. Without it, a support vector regressor reached 9.34 µg/m³ against 17.60 µg/m³ for a mean baseline, but gained little in August. Tree models given three hours of PM2.5 history reached 8.43 µg/m³ and kept skill in August (13.70 against 19.21 µg/m³). A bidirectional recurrent imputer improved gap filling but not prediction. In sparse networks, recent measurements matter more than weather inputs or sophisticated imputation.

Keywords: PM2.5 forecasting, low-cost sensors, AirQo, missing-data imputation, time-series validation, model deployment, Uganda.

1. Introduction

Fine particulate matter (PM2.5), the mass of airborne particles smaller than 2.5 µm in diameter, is the pollutant of greatest health concern in most cities, and the World Health Organization's annual guideline for it is 5 µg/m³ [1]. Reference-grade monitors are too expensive to deploy densely in most African cities. In Uganda, the AirQo network has filled part of that gap with low-cost optical sensors placed across urban areas [2], [3]. The price of that coverage is accuracy and continuity: low-cost readings have to be calibrated against reference instruments [4], they agree less well with reference monitors at high concentrations and at hourly resolution [5], and devices go offline for hours or days at a time.

Most machine-learning studies of PM2.5 forecasting work with one or more years of continuous data [8]-[11] [AQ1]. A city agency, researcher or developer working with a newly deployed or intermittently reporting network may have far less. This paper reports what we learned when we built, evaluated and deployed an hourly PM2.5 predictor on such a record: a file nominally covering January to July 2026 that, on inspection, holds readings on only eight days. It is written for practitioners and researchers who build data products on low-cost sensor networks, particularly in African cities where such networks are expanding.

The work went through two pipelines. The first used PM10 as an input and appeared highly accurate. Asking why it was so accurate showed that the model relied on a same-hour co-measurement of the target, and we rebuilt the pipeline without it. The rebuilt pipeline, and the alternatives we tested and rejected along the way, are the subject of this paper.

1.1 Objectives

The paper has four objectives:

1. to measure how much hourly PM2.5 skill a sparse record supports from weather, time and sensor identity alone, and how much short PM2.5 history adds;
2. to test whether a more accurate method for filling missing weather data improves the predictions made from it;
3. to check how stable these results are across validation designs and on a later, independent month; and
4. to show that the resulting model can be served on free cloud tiers, and to report the constraints encountered.

The contribution is empirical rather than methodological. All the methods used are existing ones. What the paper adds is evidence, for one network and one period, that weather-based models trained on a few days of data do not transfer to new periods while models with recent PM2.5 do; that an imputer which is clearly better at filling gaps need not improve downstream prediction; and that an implausibly accurate first model can be traced to its use of a co-measured pollutant.

2. Related Work

AirQo's own studies describe the conditions under which its data are collected. Okure et al. [2] characterised ambient air quality in selected Ugandan urban areas with low-cost sensing, and Sserunjogi et al. [3] published an hourly PM2.5 and PM10 dataset from the spatially distributed AirQo network [AQ2]. Data quality is a recurring theme. Adong et al. [4] applied machine learning to large-scale field calibration of AirQo PM2.5 and PM10 sensors against a collocated reference monitor [AQ3]. Lsoto et al. [5] compared 11 low-cost PM2.5 sensors with a beta attenuation monitor in Kampala over 13 months, at hourly and daily resolution, and proposed a stepwise framework for interpreting sensor data [AQ4]. The calibrated values used in our study therefore still carry sensor error, especially at hourly resolution.

A second line of work concerns meteorology. Wang and Ogawa [6] analysed the effects of meteorological conditions on PM2.5 in Nagasaki, and Yang et al. [7] found that the relationships between PM2.5 and meteorological factors across China vary by season and region. Closest to our setting, Wabinyai et al. [8] modelled daily PM2.5 in Kampala and Fort Portal with meteorological variables and lagged pollution [AQ5]. These studies motivate weather as an input, but they also show that its effect depends on place and season, which a record of a few days cannot sample.

The third line concerns forecasting models and how they are validated. Febrian and Permai [9] forecast daily PM2.5 in Central Jakarta seven days ahead from two years of pollutant and meteorological data. They compared CNN, LSTM and CNN-LSTM models under pollutant-only and pollutant-plus-meteorology inputs, and found that the LSTM with pollutant-only input gave the most consistent results; they concluded that historical pollutant patterns alone were sufficient for short-term forecasting and that meteorology added limited benefit. Niyonzima et al. [10] forecast PM2.5 one hour ahead on a large AirQo record with recurrent networks [AQ6], and Gürtepe et al. [11] predicted PM2.5 regionally with machine learning and evaluated it at stations withheld from training, a design that addresses spatial data gaps [AQ7].

For methods we drew on BRITS [12], a bidirectional recurrent imputer built from long short-term memory (LSTM) cells [13]; on SMOTE [14] and its regression variant [15] for rare high-pollution hours; and on SHAP [16], [17] for model explanations. Among the studies we examined, the forecasting work uses records of a year or more, and none asks what can be learned from a record of a few days or tests whether better imputation of the

inputs changes downstream accuracy in that regime. We make no claim beyond the literature we examined.

3. Methodology

3.1 Data

The main file contains 14,600 hourly records with 12 fields: site name, timestamp, sampling frequency, network (AirQo or AirGradient), latitude, longitude, temperature, relative humidity, calibrated PM2.5 and PM10, site identifier and device name [AQ8]. There are 149 devices at 145 named sites, so the device, not the site, identifies a sensor; no device-hour pair is duplicated. Table 1 summarises the properties that shaped the study. The most important is coverage: although the file is labelled January to July 2026, readings exist only on 1-3 January, 1-2 April and 1-3 June, 147 hours in which at least one sensor reported. About 28.8% of rows from AirQo devices report exactly 0 °C and 0% humidity together, in long blocks, and no AirGradient row does; we treat these as weather-sensor dropouts. In 22.4% of rows PM2.5 exceeded PM10, which is physically impossible because PM2.5 is a fraction of PM10, and points to calibration error [AQ9].

Table 1: Properties of the two datasets used.

Property	Main dataset	August dataset
Hourly records	14,600	73,274
Sensors	149 (AirQo and AirGradient)	125 (AirQo; all also in main data)
Days with readings	8 (1-3 Jan, 1-2 Apr, 1-3 Jun 2026)	31 (1-31 Aug 2026)
Valid PM2.5 readings	14,275	not reported; extremes not replaced
Weather zeros treated as missing	c. 28.8% of AirQo rows	c. 14% of rows
Mean PM2.5 (µg/m³)	32.7 raw; 23 in the hold-out days	35.5
Role	training and validation	external test only, no retraining

A second file covering all of August 2026 (73,274 rows, 125 sensors) became available later. It was used only as an external test set, after the same cleaning and without retraining.

3.2 Preparation and Imputation

PM10 was removed at loading. PM2.5 is the fine fraction of PM10, so the two are measured from overlapping particles, by the same instrument, in the same hour. A model given PM10 learns that overlap rather than anything that can be known in advance, and PM10 would itself have to be forecast before it could be used. Zeros in temperature, humidity, PM2.5 and coordinates were set to missing. Extreme PM2.5 readings, defined as values above the sensor's own third quartile plus three interquartile ranges, were replaced by the nearest normal reading from the same sensor within six hours, or by the sensor median if none existed. This affected 225 of 14,275 readings (1.5%) at 59 sensors; 223 were replaced by a reading at most five hours away (median one hour), and the maximum fell from 998 to 415 µg/m³. The original values were retained, and models were also scored against them. PM2.5 targets were never imputed for training or scoring.

Missing temperature and humidity were filled by one of three methods. (a) Windowed interpolation draws a straight line across a gap only when the gap is at most three hours long, has real readings on both sides and lies in a 24-hour window with at least six real readings. (b) The time-of-day method takes each sensor's mean for that hour of the day, with fall-backs to the sensor's network and then to all sensors, plus an offset estimated from the sensor's own readings within 12 hours. (c) A BRITS-style bidirectional LSTM [12], [13] reads each sensor's series forwards and backwards; each direction estimates every value before seeing it, its memory decays with the time since the last real reading, and a consistency penalty (weight 0.1) pulls the two directions together. Our version adds the

hour of day and the sensor's offline share as conditioning inputs, models PM2.5 on a log scale, hides blocks of 1-12 hours during training and drops BRITS's classification component. Every fitted component, including profiles, scalers and the imputer, was fitted on the training part of each split only. The final models use method (a) followed by method (b); Section 5.3 explains why.

The inputs are temperature, humidity, the hour of day, day of week and month (each encoded as a sine-cosine pair so that, for example, 23:00 sits next to 00:00) and a one-hot encoding of the device. We call models with only these inputs forecast models, because they can predict any future hour given the calendar and the weather. A second family adds PM2.5 history: the same sensor's real readings one, two and three hours earlier, their mean and their count. We call these nowcast models, because they need live readings.

3.3 Models and Validation

Twelve models were compared with a baseline that always predicts the training mean: linear, polynomial and ridge regression; untuned and tuned random forests [18]; support vector regression (SVR); k-nearest-neighbour regression; LightGBM [19]; an LSTM on six-hour windows of weather and time; an LSTM that also sees the previous hour's PM2.5; and random forest and LightGBM with PM2.5 history. Hyperparameters were tuned by grid or random search with time-ordered cross-validation on the training rows (three splits; two for SVR), using scikit-learn [20]. SMOTE for regression [14], [15] was tested on training rows only, at two strengths.

Because a single two-day hold-out is a thin basis for conclusions, every model was scored under the three designs in Table 2, and then on the August file. MAE was the selection metric; we also report the root mean squared error (RMSE), the coefficient of determination (R^2) and the share of hours placed in the correct air quality index category. To measure imputation accuracy, we hid real readings in the hold-out period (one gap per sensor at a random position, gaps of 1, 3, 6, 12 and 24 hours, three repeats) and compared each method's estimates with the truth. To measure the downstream effect, five models (SVR, LightGBM, LightGBM with history and both LSTMs) were retrained on weather filled by each method under all three designs. Input drift between training and evaluation data was measured with Evidently [21] (version 0.7.23), which flags a feature when its normed Wasserstein or Jensen-Shannon distance exceeds 0.1. Feature attributions were computed with SHAP [16], [17] on 400 test rows.

Table 2: The three validation designs.

Design	Training and test data
Chronological hold-out	Train 1 Jan-1 Jun (11,633 rows); test 2-3 Jun (2,642 rows, 28 hours)
Expanding-window cross-validation	Test on 1 Apr, 2 Apr and 1 Jun in turn, training on all earlier data
Leave-sensors-out	Three folds, each withholding about 50 sensors entirely

4. Technology Description

The pipeline runs as a single notebook, from loading to monitoring, without manual steps. Its recorded run on a local machine (Windows 11, Python 3.13, 16 logical CPUs, no GPU) executed 346 code cells without errors in about 56 minutes, with a peak memory use of 1.23 GB [AQ10]. The selected forecast model is serialised with its preprocessing and target scaling into a single 1.2 MB file, and a standalone inference script reproduced the notebook's predictions on all 2,642 test rows with zero difference.

Deployment took four attempts, and the failures are informative for low-budget projects. A GPU rental service required prepaid credit that the project could not afford. An attempt to run the notebook on a free hosted notebook platform did not complete. A model-hosting platform accepted the model and a static page of stored forecasts, but refused

server-side apps without a paid plan (HTTP 402), so the page could not run the model. The final deployment splits the problem in two: a lightweight Python web service on a free Render instance loads the forecast model and answers prediction requests, and a dashboard hosted on Vercel calls the service from the user's browser [AQ11]. The service exposes a health check and a prediction endpoint; each request row carries temperature, humidity, the cyclical time features and the sensor identifier, and the response gives predicted PM2.5 and its index category [AQ12]. Only the forecast model is served, because the nowcast model would need each sensor's readings from the previous three hours at request time. On the local machine a single prediction took about 14 ms and 3,576 rows about 1.4 s; latency on the hosted service was not measured.

5. Results

5.1 A Misleadingly Accurate First Model

The first pipeline used PM10 as an input and the same chronological split. Its tuned random forest reached an MAE of 3.10 µg/m³, an RMSE of 7.00 µg/m³ and $R^2 = 0.898$, with 86.2% of test hours in the correct index category. PM10 carried about 76% of the model's feature importance; temperature and humidity carried about 1% each. When the model was used to forecast August, PM10 had to be replaced by each sensor's typical value for the hour, and the forecasts simply repeated each site's usual daily pattern. The accuracy came from a co-measurement of the target, not from predictive information, and the remaining results all refer to the rebuilt pipeline without PM10.

5.2 Weather-Only Models against Models with PM2.5 History

Table 3 and Figure 1 summarise the main comparison. On the chronological hold-out, the tuned SVR was the best forecast model (MAE 9.34 µg/m³, $R^2 = 0.34$), well ahead of the mean baseline (17.60 µg/m³). Adding three hours of PM2.5 history gave the best results overall: LightGBM with history reached 8.43 µg/m³ ($R^2 = 0.42$), and history lowered the hold-out MAE of the random forest by 11.6% and of LightGBM by 14.2%.

Table 3: MAE (µg/m³) of selected models under each evaluation; hold-out R^2 in brackets. CV: expanding-window cross-validation.

Model	Hold-out	CV	Leave-sensors-out	August 2026
LightGBM + PM2.5 history	8.43 (0.42)	12.08	9.49	14.02
Random forest + PM2.5 history	8.50 (0.41)	10.25	9.03	13.70
SVR, tuned (forecast model)	9.34 (0.34)	14.68	15.09	18.32
Random forest, untuned	9.61 (0.25)	27.02	14.90	19.05
LightGBM, tuned	9.82 (0.25)	27.45	15.13	18.79
KNN, tuned	10.01 (0.26)	16.69	15.60	18.65
LSTM + PM2.5 history	10.89 (0.30)	12.18	9.41	19.31
LSTM, weather and time	12.22 (0.01)	20.70	15.61	19.47
Baseline (training mean)	17.60 (-0.30)	24.11	20.33	19.21
Ridge regression, tuned	22.00 (-0.86)	23.87	16.42	43.14

The other designs separate the two families sharply. In expanding-window cross-validation the tree-based forecast models were worse than the baseline (about 27 against 24.11 µg/m³), while the same models with history stayed near 10-12 µg/m³. At sensors withheld from training, the history models reached about 9.0-9.5 µg/m³ (R^2 of 0.63-0.65), against about 15 µg/m³ for the forecast models. Index category accuracy is a weak yardstick on these data: the SVR placed 75.2% of hold-out hours in the right category, but always predicting the training mean already scored 74.4%, because most hours fall in the Moderate band. Scored

against the untouched readings rather than the cleaned ones, the hold-out MAE was 9.79 µg/m³ for LightGBM with history and 10.62 µg/m³ for the SVR.

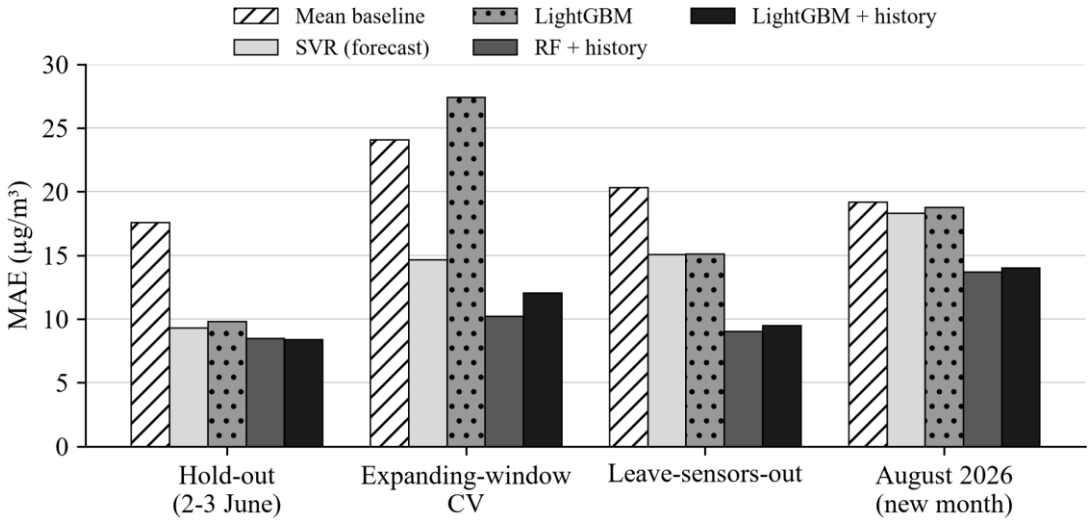


Figure 1: MAE of the mean baseline, two forecast models and two models with PM2.5 history under the three validation designs and on the August 2026 dataset.

SHAP attributions show why. Without history, the random forest and LightGBM placed 46% and 32% of their attribution on the month and 18% and 27% on sensor identity, but only 5-8% each on temperature and humidity. With six training days spread over three months, the month mostly tells a model which burst a row came from. With history, half of LightGBM's attribution moved to the recent readings. Errors also depended on data availability: the SVR's hold-out MAE was 8.73, 10.06 and 10.76 µg/m³ for data-rich, intermediate and data-poor sensors, and LightGBM with history 7.66, 9.22 and 10.42 µg/m³; history helps least where a sensor is often offline. Neither family reproduced peaks: the highest test reading, 172 µg/m³, was predicted as 33 µg/m³.

5.3 Better Imputation, Unchanged Predictions

As an imputer, the bidirectional model was clearly better than the time-of-day method. Its error was lower in all 15 combinations of variable and gap length, by 20.8% on average (Table 4). Interpolation was most accurate for one- to three-hour weather gaps, but it could fill only 618 of the 11,423 missing temperature values. The bidirectional model's fills were smoother than real data: their standard deviation for PM2.5 was 11.9 µg/m³ against 26.8 µg/m³ observed.

Table 4: Imputation MAE on hidden hold-out values, by gap length. (a) does not fill gaps longer than three hours.

Variable	Method	1 h	6 h	24 h
Temperature (°C)	(a) / (b) / (c)	0.49 / 1.61 / 0.76	- / 1.87 / 1.36	- / 2.13 / 1.97
Humidity (%)	(a) / (b) / (c)	1.80 / 5.91 / 2.90	- / 6.76 / 5.66	- / 7.99 / 7.38
PM2.5 (µg/m³)	(a) / (b) / (c)	7.35 / 9.10 / 6.85	- / 10.42 / 9.80	- / 12.13 / 10.26

The downstream test gave a different picture. Training on the bidirectional imputer's weather instead of the time-of-day weather changed prediction MAE by -0.4% on average and helped in only 7 of 15 model-design cases. It helped both LSTMs on the hold-out (from 10.89 to 9.48 µg/m³ with history), hurt the SVR in cross-validation (from 14.68 to 16.35 µg/m³) and changed the rest by about 1% or less. The final models therefore keep methods (a) and (b). The decision rests on five models; the others were not retested.

5.4 Oversampling Rare High-Pollution Hours

SMOTE for regression made all eight models on which it was tried worse on the hold-out, at both strengths; at a strength of 0.25, for example, LightGBM rose from 9.82 to 28.67 µg/m³ and the untuned random forest from 9.61 to 10.10 µg/m³. Oversampling high-pollution hours raises the average level a model learns, and nothing in the inputs tells it where those hours belong, while the test days were cleaner than the training period (mean 23 against 33 µg/m³). The final models use real rows only.

5.5 Transfer to a New Month

August was more polluted than the hold-out days (mean 35.5 against 23 µg/m³). On it, the forecast models were barely better than the baseline (SVR 18.32 against 19.21 µg/m³), all with R^2 below zero, whereas the random forest with history reached 13.70 µg/m³ (R^2 = 0.30). Aggregated to daily means, the forecast model predicted the Moderate category on all 31 days, while the observed mean fell in the Unhealthy for Sensitive Groups band on 16; it matched the observed category on 15 days and the history models on 17. Unfamiliar sensors do not explain this, because every August sensor was in the training data, and missingness does not either, since 76 of the 125 sensors were data-rich in August. Evidently flagged six of nine input features as drifted between training and August, most strongly the month (0.83), day of week (0.75 and 0.61), humidity (0.26) and the mix of sensors (0.25). Between training and the hold-out, eight of nine features had already drifted, largely because the test window holds two weekdays of one month. The imputer, by contrast, transferred well: on August temperature its six-hour error was 1.3 °C.

6. Discussion

The results tell a consistent story. With a few days of data, a model without PM2.5 history can only learn the level of each burst and each sensor, and the month feature acts as a label for the burst. That knowledge does not carry over to another period, which is why the forecast models fail in expanding-window cross-validation and in August. Three hours of recent PM2.5 tells the model the current level directly, which is why the tree models with history are the most stable across every design we used. The LSTM with history is the exception: it did well in the three designs but not in August, and we cannot yet say why [AQ16]. Weather mattered little here; this is a statement about what eight days can teach a model, not about the physical role of weather, which [6]-[8] document over far longer records.

The direction of these findings agrees with prior work, with caveats. Febrian and Permai [9] reached a similar conclusion about pollutant history for daily forecasts in Jakarta, from two years of data and a different model family. Wabinyai et al. [8] likewise found lagged PM2.5 useful in Kampala [AQ5]. Our best hold-out MAE is of the same order as the one-hour-ahead error reported by Niyonzima et al. [10] for a much larger AirQo record [AQ6], but the data volumes, horizons and validation designs differ too much for a direct comparison. As in Gürtepe et al. [11], prediction at unseen stations was harder than prediction at known ones for most of our models [AQ7].

Two findings concern how such studies are run rather than which model wins. First, an excellent score deserves suspicion: the PM10 model's 3.10 µg/m³ would have been reported as a success had we not asked what it relied on. Second, imputation should be judged by its downstream effect. The bidirectional imputer was the best gap-filler we built, yet it did not change the deployed model, partly because weather carries so little of the attribution. Its value would rise if weather became a stronger input or if the imputed series were itself the product.

The demonstrated impact is modest and should be stated as such: a reproducible pipeline and a working prediction service with a dashboard, built and hosted at no cost. No users have evaluated the dashboard, and no agency has used the forecasts. The expected impact lies in practice for teams working with new or intermittent networks: evaluate models under more than one time-aware design and on an out-of-period test, treat co-measured pollutants as a leakage risk for forecasting, prefer nowcasting where live readings exist, and invest in continuous data before complex imputation.

The limitations are substantial. Eight days in three bursts cannot reveal seasonal patterns, and the hold-out covers only 28 hours. The data come from one network in one country, and the calibrated readings still contain sensor error. Replacing extreme values lowers the reported errors (by about 1.3 µg/m³ in hold-out MAE against the untouched readings), and the replacement was not applied to the August file. The imputation decision was tested on five models only. We report no significance tests or run-to-run variance [AQ13], and latency was measured locally, not on the hosted service.

7. Conclusions and Summary Recommendations

We built, evaluated and deployed an hourly PM2.5 predictor on an AirQo record containing eight days of readings. A first model looked excellent because it used PM10, a same-hour co-measurement of the target. Without it, a tuned SVR forecast model reached a hold-out MAE of 9.34 µg/m³, but it and the other weather-only models did not transfer to new periods. Random forest and LightGBM models with three hours of PM2.5 history reached 8.43-8.50 µg/m³, were the most stable under every validation design and kept part of their skill on a later, full month. A BRITS-style imputer improved gap filling by about a fifth but changed prediction error by less than half a percent, and SMOTE oversampling made every model worse.

For teams in similar settings we recommend validating with time-aware and out-of-period tests rather than a single split, excluding co-measured pollutants from forecasting inputs, serving nowcasts where live readings are available, and judging imputation by its effect on the final task. Further work should retrain on continuous data such as the August file, combine lagged PM2.5 with multi-step forecasting and forecast weather, use readings from neighbouring sensors, and automate drift monitoring and retraining.

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Claude (Anthropic), used through the Claude Code tool, produced a first draft of the text, Tables 1-4 and Figure 1 of this paper from the authors' project report, analysis notebook outputs and cited sources, and checked numerical values and reference metadata against those materials. The authors reviewed, revised and verified the content and take full responsibility for it. [AQ15: The authors must confirm and complete this declaration, including any AI assistance used in developing the analysis code or the original project report, and must revise the draft text themselves before submission.]

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